



**UNIVERSITY OF
CALGARY**

Predicting Daily PM_{2.5} in Calgary, Canada Using Weather Data- Final Report

Course: ENGG 680 L01 – Introduction to Digital Engineering, FALL 2025

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Group Number: 23

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Certificate of Work

Title of Assignment: Project Report on Predicting Daily PM2.5 in Calgary, Canada Using Weather Data

Group Number: 23

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We request that the total mark for this assignment be distributed as follows among group members:

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Total Contribution: 100%

1.Introduction

1.1 Background on PM2.5

Fine particulate matter (PM2.5) refers to airborne particles with a diameter of 2.5 micrometers or smaller. Due to their small size, these particles can penetrate deep into the respiratory system, reach the bloodstream, and contribute to a range of adverse health impacts including asthma, cardiovascular disease, lung inflammation, and premature mortality. PM2.5 originates from both natural and human-made sources such as wildfire smoke, industrial emissions, vehicle exhaust, construction activities, and secondary aerosol formation. In Western Canada, increasingly severe wildfire seasons have resulted in extended episodes of elevated PM2.5 that pose significant public health and environmental challenges.

1.2 Importance of Forecasting PM2.5

Accurate PM2.5 forecasting is essential for informing early-warning systems, supporting public health decision-making, and guiding municipal air-quality management strategies. Predictive models allow government agencies and local authorities to anticipate periods of poor air quality and issue timely advisories. From an engineering standpoint, PM2.5 forecasting plays a critical role in environmental monitoring, urban planning, transportation management, and climate-resilience initiatives. A dependable forecasting tool helps minimize exposure risks for vulnerable populations and supports evidence-based environmental policy.

1.3 Literature Review

Air pollution is widely recognized as a major global health concern, with fine particulate matter (PM2.5) identified as one of its most harmful components due to its ability to penetrate deep into the respiratory and cardiovascular systems (World Health Organization, 2023). Extensive epidemiological evidence shows that exposure to elevated PM2.5 levels increases the risk of respiratory and cardiovascular disease, and even relatively small day-to-day increases in PM2.5 concentrations have been linked to higher rates of hospital admissions among older adults and individuals with pre-existing chronic conditions (Brook et al., 2010).

In response to these risks, a substantial body of research has focused on developing predictive models capable of forecasting short-term PM2.5 levels. Deep learning approaches such as Long Short-Term Memory (LSTM) networks have demonstrated strong performance in capturing temporal dependencies in air pollutant data, particularly when meteorological and environmental predictors are incorporated (Li et al., 2017). Within the Canadian context, machine learning models that integrate satellite-based data with ground-level environmental variables have shown high accuracy in estimating PM2.5 concentrations at fine spatial and temporal scales (Liang et al., 2021). Ensemble learning methods, including Random Forest and Gradient Boosting, have also been widely used due to their capacity to model nonlinear interactions between weather variables and pollutant concentrations (Zhang et al., 2012). Broader epidemiological and air-quality modeling research highlights the long-term health impacts of PM2.5 and underscores the importance of developing reliable pollutant estimation frameworks (Burnett et al., 2018; Kloog et al., 2014).

Despite substantial progress, several limitations remain in current forecasting approaches. Many models rely on dense monitoring networks, satellite products requiring specialized processing, or region-specific calibration parameters, making them difficult to generalize to smaller urban areas or locations with fewer monitoring stations. Prior research also points to challenges such as

missing meteorological observations, data imbalance during extreme pollution events, and the need to adapt models to local climatic conditions. These gaps motivate the current project, which aims to develop a streamlined, locally calibrated machine learning model, using readily accessible meteorological variables, to generate reliable daily PM2.5 forecasts for Calgary.

1.4 Existing Solutions and Their Limitations

Current PM2.5 forecasting solutions in Canada primarily rely on large-scale numerical weather prediction (NWP) models and chemical transport models (CTMs). Although these systems provide physically grounded simulations, they require substantial computational resources, operate at coarse spatial resolutions, and often fail to respond quickly to sudden wildfire smoke intrusions or rapidly changing weather conditions. Traditional statistical models used in some public advisory systems assume linearity between environmental variables and PM2.5, making them less effective when dealing with the highly non-linear and event-driven nature of air pollution.

Machine-learning approaches proposed in research studies demonstrate improved predictive performance, but many require extensive datasets, specialized features, or satellite inputs that may not be practical for streamlined, city-level applications. Moreover, few existing models have been explicitly calibrated for Calgary's unique meteorological conditions and wildfire exposure patterns.

1.5 Gaps Addressed by This Project

This project addresses the following key gaps in existing PM2.5 forecasting approaches:

- Limited availability of simple, data-driven forecasting tools specifically tailored for Calgary
- Insufficient use of autoregressive temporal signals despite their strong predictive value
- Underutilization of basic meteorological variables that can enhance predictive power with low data complexity
- Lack of lightweight, locally calibrated machine-learning models suitable for operational forecasting
- Need for models that remain resilient during high-PM2.5 wildfire events

1.6 Engineering Relevance

Air-quality forecasting is directly connected to environmental engineering, urban systems design, and public-health risk mitigation. Reliable PM2.5 prediction supports:

- urban planning and zoning
- traffic and transportation management
- emissions control strategies
- emergency preparedness during wildfire seasons
- regulatory compliance and environmental reporting

By creating an accurate and efficient prediction model, this project contributes to the development of practical engineering tools that can enhance Calgary's environmental resilience.

1.7 Project Statement

The objective of this project is to develop and evaluate a machine-learning-based forecasting system that predicts next-day PM2.5 concentrations in Calgary by integrating historical pollutant

trends with key meteorological variables to create an accurate, lightweight, and locally calibrated air-quality prediction tool.

2. Dataset Description

This section provides an overview of the datasets used in this study, including the structure, variables, and temporal attributes that form the foundation for subsequent analyses and modeling. The final dataset consists of daily PM2.5 measurements and meteorological observations for the City of Calgary.

2.1 Overview of the Dataset

The dataset integrates daily air-quality and weather information and covers a continuous multi-year period. All observations are aligned at a daily resolution, resulting in a complete, gap-free time series after preprocessing. This dataset is well-suited for short-term PM2.5 forecasting because it includes variables known to influence atmospheric pollution levels. The final cleaned dataset contains 1461 daily observations covering the full study period.

2.2 PM2.5 Variable

The primary variable of interest is daily mean PM2.5 concentration (measured in $\mu\text{g}/\text{m}^3$). These values represent the average fine particulate matter concentration recorded across the 24-hour period for each date. PM2.5 is a key indicator of air quality, strongly affected by wildfire smoke, atmospheric mixing conditions, and urban emissions.

2.3 Meteorological Variables

The dataset includes a range of daily meteorological factors obtained from environmental monitoring stations. These variables are essential for understanding atmospheric behavior and pollutant dispersion. They include:

- Mean temperature ($^{\circ}\text{C}$)
- Maximum and minimum temperature ($^{\circ}\text{C}$)
- Relative humidity (%)
- Total precipitation (mm)
- Wind speed (km/h)
- Wind direction (degrees)
- Atmospheric pressure (kPa)

Each variable influences PM2.5 concentrations through mechanisms such as dilution, vertical mixing, or precipitation scavenging.

2.4 Temporal Features

To capture seasonal and weekly atmospheric dynamics, several temporal variables were derived from the date field:

- month (1–12), reflecting seasonal cycles
- dow (day of week; 0–6), representing potential behavioral patterns in emissions
- season, categorizing observations into winter, spring, summer, and fall

These features provide contextual information related to seasonal trends and human activity cycles.

2.5 Final Variable List

The final dataset used for modeling includes the following variables:

- PM2.5 (target variable)
- Mean temperature
- Total precipitation
- PM25_lag1
- PM25_roll3
- month
- dow
- season

This structured set of variables captures environmental conditions, temporal patterns, and short-term pollutant persistence relevant for forecasting.

3. Data Collection and Integration

This section describes how the raw datasets were collected, processed, and merged to form the unified daily dataset used for analysis and modeling. It outlines the data sources, retrieval steps, and integration procedures applied before formal cleaning.

3.1 Data Sources

Two primary open-data sources were used:

- PM2.5 data from the National Air Pollution Surveillance (NAPS) program
- Meteorological data from Environment and Climate Change Canada (ECCC) via the Historical Climate Data portal

Both sources provide regulated, validated, and publicly accessible environmental measurements suitable for scientific analysis.

3.2 Data Retrieval Procedures

Raw datasets were downloaded directly from their respective portals. PM2.5 data were originally provided at an hourly resolution, while the meteorological dataset contained daily values. To match temporal scales, PM2.5 observations were aggregated into daily averages. All data files were imported into the analysis environment in their native CSV format.

3.3 Data Alignment and Merging

To integrate the two datasets:

- The date field (YYYY-MM-DD) was used as the common key
- Hourly PM2.5 values were aggregated into daily means before merging
- An inner join was applied to retain only dates present in both datasets
- Records were sorted chronologically to create a continuous timeline

This produced a synchronized dataset with one row per date and complete coverage of all variables.

3.4 Handling Initial Inconsistencies

Before applying formal cleaning steps, several inconsistencies were addressed:

- Dates missing in one dataset but present in the other

- Occasional missing hourly PM2.5 values prior to aggregation
- Isolated inconsistencies in weather station readings, such as absent wind-direction values

These issues were identified during initial inspection and corrected to ensure consistency prior to detailed cleaning.

3.5 Final Integrated Dataset

After aggregation and merging, the resulting dataset consisted of aligned daily observations for all selected PM2.5 and meteorological variables. This integrated dataset served as the input for the data-cleaning procedures described in Section 4 and formed the basis for exploratory analysis and modeling.

4. Data Cleaning

A structured, multi-step data cleaning pipeline was implemented to ensure accuracy, consistency, and completeness of the final dataset. All preprocessing steps were applied to the integrated daily dataset and resulted in the production of the final cleaned file:

Calgary_PM25_Weather_Cleaned_v3.csv

4.1 Handling Missing and Incomplete Records

Both datasets contained missing timestamps and incomplete observations. The following procedures were applied:

- Hourly PM2.5 and weather readings were aggregated into daily values to reduce high-frequency gaps.
- Remaining gaps in continuous weather variables were filled using linear interpolation, preserving natural trends.
- Missing precipitation values were replaced with zero, as this is physically meaningful.
- Days with insufficient PM2.5 coverage after aggregation were removed to avoid biased pollutant estimates.

These steps produced a complete and continuous daily time series suitable for temporal modeling.

4.2 Removing Duplicates and Inconsistent Values

- Duplicate dates introduced during merging were identified and removed.
- Inconsistent or impossible values were corrected or discarded after manual verification.

4.3 Outlier Detection and Correction

To prevent extreme observations from disproportionately influencing the model, an IQR-based outlier filter was applied separately to:

- PM2.5
- Temperature
- Relative humidity
- Wind speed

Values falling beyond $\pm 1.5 \times \text{IQR}$ were removed and replaced using short-range interpolation, preserving realistic variability while reducing noise.

4.4 Temporal Alignment and Indexing

- All observations were standardized to the YYYY-MM-DD format.
- PM2.5 and weather datasets were merged using an inner join, ensuring complete alignment.
- The merged dataset was sorted chronologically and re-indexed to maintain a clean temporal structure.

This ensures that each row corresponds to a complete daily observation for Calgary.

4.5 Preparing the Final Clean Dataset

Following the full cleaning workflow:

- No missing values remained
- All predictor variables were numeric
- The dataset was chronologically consistent
- PM2.5 retained natural variability suitable for lag-based modeling

This final dataset served as the foundation for EDA, feature engineering, and model development.

4.6 Python Libraries Used for Preprocessing

All preprocessing steps in this study were implemented using widely adopted Python libraries. Pandas was used for data ingestion, data cleaning, interpolation of missing values, and date-based indexing. NumPy supported numerical operations and array manipulation during aggregation and feature creation. Scikit-learn provided tools for generating lagged and rolling features, splitting the dataset into training and testing subsets, and preparing the feature matrix for model development. These libraries form a standard and reliable toolkit for data preprocessing and machine learning workflows in Python.

5. Exploratory Data Analysis (EDA)

The Exploratory Data Analysis (EDA) phase was conducted to examine the temporal structure, seasonal behavior, and statistical characteristics of PM2.5 concentrations in Calgary. This step validates the quality of the cleaned dataset and ensures that the updated preprocessing pipeline preserves real environmental patterns while removing noise, gaps, and inconsistencies.

5.1 Temporal Trends

Daily PM2.5 concentrations were plotted across the study period to investigate long-term patterns and overall variability. The time-series visualization showed:

- Recurring seasonal fluctuations, with peaks typically occurring in late summer and early fall
- Elevated values during wildfire smoke episodes associated with Western Canada's fire season
- Moderately higher baseline levels in winter, likely due to reduced vertical mixing and atmospheric inversions

The updated cleaning pipeline (interpolation + outlier filtering) preserved the natural structure of the PM2.5 time series without distorting its temporal behavior.

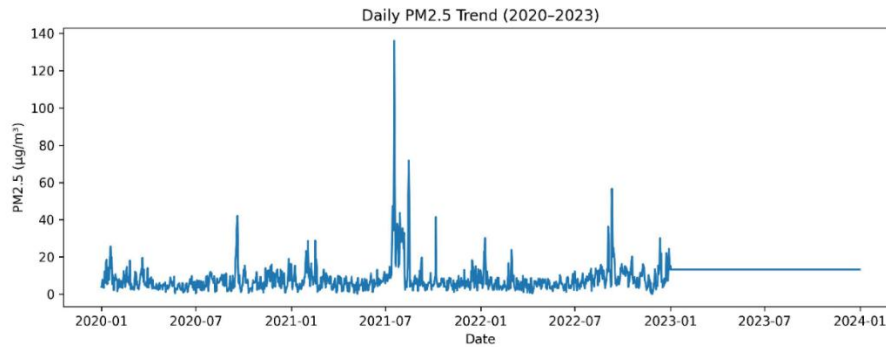


Figure 5.1: Daily PM2.5 levels in Calgary (2020–2023)

5.2 Seasonal and Monthly Patterns

Monthly and seasonal aggregations revealed consistent and physically meaningful patterns:

- Summer and early fall exhibited the highest PM2.5 concentrations due to wildfire smoke and stagnant warm-season air masses
- Spring maintained the lowest and most stable values
- Winter showed moderate increases related to reduced dispersion and cold-weather emissions

These patterns justify the inclusion of month and season as predictive features.

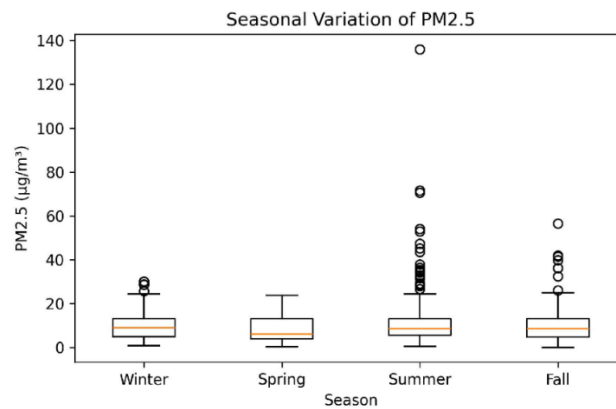


Figure 5.2: Seasonal variation of PM2.5 levels in Calgary

5.3 Weekly Structure

A day-of-week (dow) analysis revealed mild but consistent differences:

- Slightly higher concentrations on weekdays
- Lower levels on weekends

These trends align with typical human activity patterns and support the use of dow as a temporal feature.

The cleaned dataset's continuity ensures these patterns were not artificially distorted by missing data.

5.4 Correlation Analysis

A correlation matrix quantified linear relationships between PM2.5 and meteorological variables. Key findings:

- Negative correlation with temperature, indicating better dispersion during warm conditions

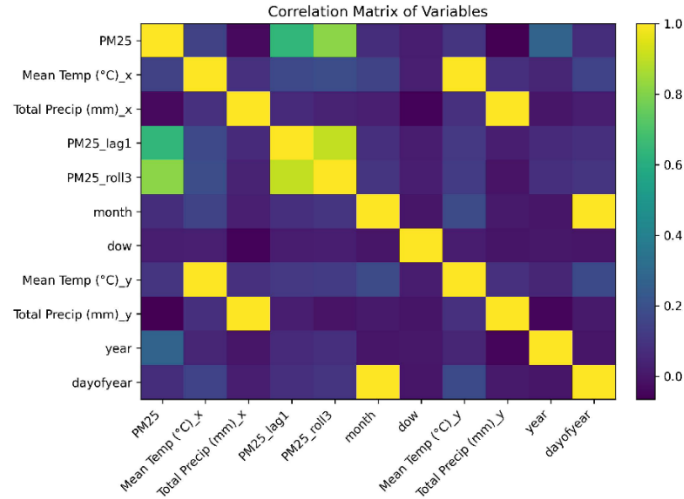


Figure 5.3: Correlation matrix of PM2.5 and meteorological variables

- Inverse relationship with precipitation, reflecting rainfall's scavenging effect

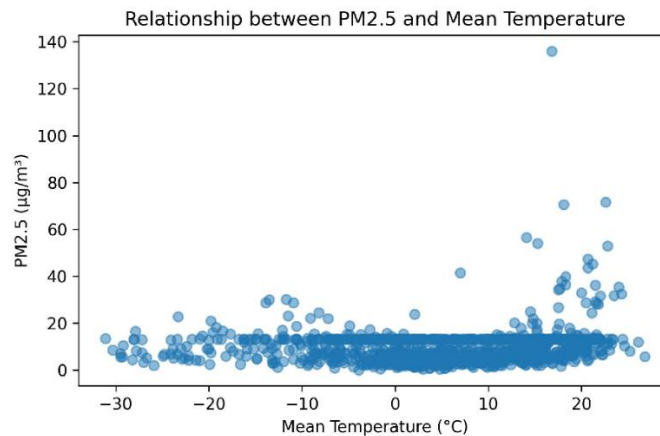


Figure 5.4: Relationship between PM2.5 and mean temperature

- Strong positive correlations with autoregressive features (PM2.5_lag1 and PM2.5_roll3), confirming the temporal dependence of PM2.5

5.5 Outlier Identification

EDA identified several genuine high-PM2.5 events associated with wildfire smoke.

The IQR-based outlier filter removed only unrealistic or sensor-error spikes while preserving authentic extreme values.

These retained peaks are essential for training a forecasting model that performs reliably during high-risk pollution episodes.

5.6 Summary of EDA Findings

The EDA confirmed that:

- PM2.5 exhibits strong temporal and seasonal structure
- The dataset shows substantial autoregressive behavior
- Meteorological variables influence PM2.5 in non-linear ways
- The updated cleaning pipeline produced a gap-free, realistic time series
- The dataset is well-suited for regression modeling and machine-learning techniques

These insights directly guided subsequent decisions on feature engineering and model selection.

6. Modeling Methodology

The modeling stage aimed to develop, train, and compare predictive models for daily PM2.5 forecasting in Calgary. Two models were implemented:

1. Linear Regression (baseline)
2. Random Forest Regressor (improved model)

Both models used the cleaned dataset: `Calgary_PM25_Weather_Cleaned_v3.csv`

6.1 Feature Selection

The final set of predictors included:

- Mean temperature (°C)
- Total precipitation (mm)
- PM25_lag1 (previous day's PM2.5)
- PM25_roll3 (3-day rolling average)
- month
- dow
- season

Target variable:

- PM25 (daily average PM2.5)

These features capture the primary temporal, meteorological, and autoregressive drivers of PM2.5 dynamics.

6.2 Baseline Model: Linear Regression

The Linear Regression model served as a transparent and interpretable benchmark.

Motivations:

- Establish a reference for minimum acceptable performance
- Provide interpretability and simplicity
- Allow clear comparison with more flexible models

The model was trained using ordinary least squares (OLS) without regularization or feature scaling.

6.3 Non-Linear Model: Random Forest Regressor

A Random Forest Regressor was implemented to evaluate whether a non-linear machine-learning approach could enhance next-day PM2.5 forecasting. Unlike Linear Regression, which assumes a strictly linear relationship between the predictors and the target variable, Random Forest can capture complex interactions, non-linear behaviors, and short-term variability that may exist in atmospheric data.

- `n_estimators = 300` to ensure stable ensemble averaging
- `max_depth = None` to allow full tree expansion
- `random_state = 42` to ensure reproducible results
- `n_jobs = -1` to enable full parallel CPU processing

The Random Forest model was included not as a guaranteed improvement over the linear baseline, but to examine whether non-linearity within the predictors provides any additional predictive advantage. Its flexibility, resistance to noise, and ability to estimate feature importance make it a valuable complementary model for assessing potential non-linear structures in the dataset.

6.4 Train/Test Split

An 80/20 split was used:

- Provides sufficient data for training
- Ensures temporal consistency with lag-based features
- Maintains reproducibility through `random_state=42`

6.5 Model Training

Linear Regression

- Trained using OLS
- Offers interpretability but limited flexibility

Random Forest

- Trained using an ensemble of 300 decision trees
- Captures non-linearities and short-term temporal dependencies
- Generates feature-importance rankings

Random Forest does not require feature scaling.

6.6 Evaluation Metrics

Both models were evaluated using:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- Coefficient of Determination (R^2)

These metrics quantify average error, penalize large deviations, and measure explained variance.

6.7 Diagnostic Checks

To evaluate the predictive behavior of the improved Random Forest model, two diagnostic procedures were defined:

1. Regression Error Analysis :to examine the distribution of prediction errors across the full range of observed PM2.5 values.
2. Residual Analysis :to assess variance stability, potential bias, and the presence of systematic patterns in model errors.

These diagnostic methods are standard in regression modeling and help determine whether the model exhibits heteroscedasticity, structural bias, or error clustering, particularly during extreme PM2.5 conditions such as wildfire-driven pollution events.

7. Model Evaluation

This section evaluates the predictive performance of both the baseline Linear Regression model and the non-linear Random Forest Regressor. The assessment includes quantitative test-set metrics and diagnostic observations regarding each model's behavior.

7.1 Baseline Model: Linear Regression

The Linear Regression model served as a simple, interpretable baseline for comparison.

Test-set performance:

- MAE: 1.141
- RMSE: 2.068
- R^2 : 0.7969

These results show that the linear model captures most of the variance in daily PM2.5 concentrations and performs consistently across a wide range of conditions. The strong performance reflects the usefulness of autoregressive features (lag and rolling averages), which enhance linear structure within the dataset and allow the model to provide accurate next-day forecasts despite its simplicity.

7.2 Non-Linear Model: Random Forest Regressor

The Random Forest model was evaluated to determine whether non-linear relationships among predictors could improve forecasting performance.

Test-set performance:

- MAE: 1.152
- RMSE: 2.235
- R^2 : 0.7627

The Random Forest model performs well overall, but it does **not** outperform the linear baseline on this dataset. This suggests that, after cleaning and feature engineering, the predictors exhibit predominantly linear behavior, with strong short-term persistence that is effectively captured by

linear methods. While Random Forest can model non-linear interactions and remains a robust alternative, it does not provide additional predictive benefit in this specific context.

7.3 Visual Assessment of Model Performance

In addition to numerical metrics, several diagnostic plots were generated to examine prediction error patterns, potential bias, and the overall predictive alignment of the modeling approach. These visual assessments provide insight into how the model behaves across different PM2.5 ranges and help identify issues such as systematic over- or under-prediction or heteroscedasticity.

7.3.1 Error Plot (Actual PM2.5 vs. Error)



Figure 7.1: Error Plot (Actual PM2.5 vs. Prediction Error)

This plot shows how prediction errors vary across the observed PM2.5 range.

Key observations:

- Errors are tightly concentrated around zero for most days.
- Larger deviations occur primarily during extreme wildfire-related events.
- No upward or downward trend is present, indicating minimal systematic bias.

7.3.2 Residual Plot (Predicted PM2.5 vs. Residuals)

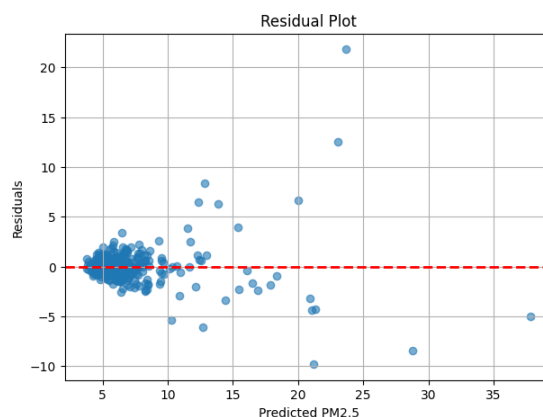


Figure 7.2: Residual Plot (Predicted PM2.5 vs. Residuals)

This plot evaluates the variance and distribution of residuals.

Key observations:

- Residuals are randomly scattered around the zero line, which indicates that the model does not exhibit directional bias.
- No funnel-shaped pattern is present, suggesting that the variance of errors remains relatively stable across the prediction range.
- A slight increase in variance at higher predicted values is expected, as extreme pollution events are rare and naturally more difficult to model

7.3.3 Actual vs. Predicted Scatter Plot

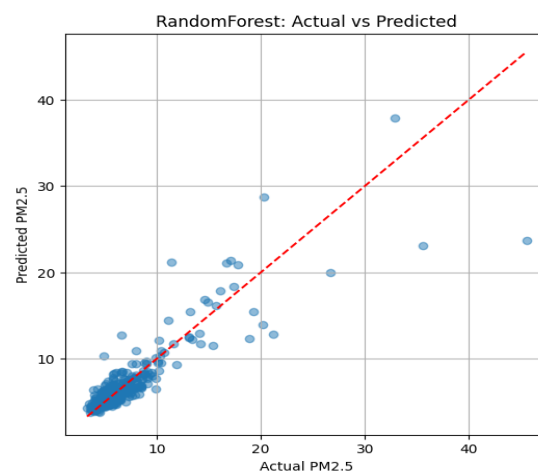


Figure 7.3: Actual vs. Predicted PM2.5

This scatter plot compares the model's predicted PM2.5 values with the actual observations.

Interpretation:

- Points are tightly clustered around the 1:1 reference line, indicating strong agreement between predictions and real measurements.
- Larger deviations at higher PM2.5 levels correspond to wildfire-related peaks, which are naturally more difficult for any model to capture accurately.
- Overall, the plot demonstrates that the model produces accurate predictions across most of the PM2.5 range

7.4 Summary of Model Evaluation

Overall, the evaluation shows that:

- The Linear Regression model slightly outperforms the Random Forest Regressor on the final cleaned dataset.
- Diagnostic plots confirm low bias, stable residual variance, and consistent predictive behavior across most PM2.5 levels.

- Both models perform reliably under typical conditions, although extreme wildfire-related peaks remain more challenging to capture.
- These findings suggest that, for this dataset, a simple linear model is sufficient and even preferable, while the Random Forest model remains a strong non-linear alternative

8. Model Comparison

This section presents a direct quantitative and qualitative comparison between the baseline Linear Regression model and the non-linear Random Forest Regressor.

8.1 Quantitative Comparison

Performance Summary

Model	MAE	RMSE	R ²
Linear Regression	1.141	2.068	0.7969
Random Forest	1.152	2.235	0.7627

Interpretation:

- Linear Regression achieves slightly lower MAE and RMSE than the Random Forest model, indicating marginally more accurate predictions.
- R² is also higher for the Linear Regression model, meaning it explains a greater portion of the variance in PM2.5 levels.
- These results suggest that the cleaned dataset exhibits mostly linear behavior, especially due to the strong autoregressive features.
- While Random Forest performs well, non-linear modeling does not provide additional predictive benefits for this dataset

8.2 Qualitative Comparison

linear Regression

- Assumes linear relationships between predictors and PM2.5
- Performs consistently across most pollution levels
- Residuals show no strong structural patterns, indicating stable model behavior
- Simple, interpretable, and well-suited to datasets with strong autoregressive structure

Random Forest

- Capable of capturing non-linear interactions among features
- Performs well overall but does not outperform the linear baseline in this dataset
- Residuals remain randomly distributed, showing no major bias
- More robust to noise and extreme events, though its flexibility does not translate into better accuracy here

8.3 Feature Importance Perspective

The Random Forest model provides insight into the relative contribution of each feature to the predictive task. Figure 8.1 presents the feature-importance scores derived from the trained

model. These scores illustrate how strongly each predictor influences the model's decision-making process.

The results show that:

- PM25_rol13 is the most influential feature.
- PM25_lag1 closely follows, confirming the strong autoregressive behavior of PM2.5 in Calgary.
- Mean temperature and total precipitation contribute smaller but meaningful effects related to atmospheric dispersion and scavenging.
- Day of week, month, and season add contextual information reflecting recurring activity patterns and seasonal cycles.

These importance rankings are consistent with the environmental and temporal patterns observed in the EDA and confirm that the selected feature set captures the key drivers of PM2.5 variability, regardless of which model ultimately performs best

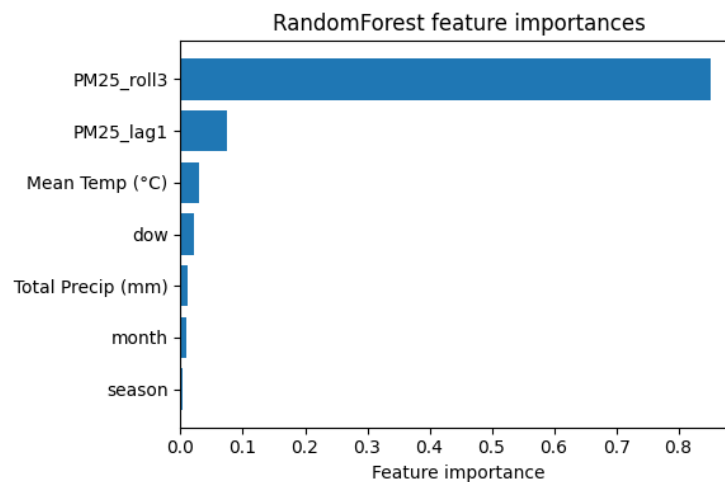


Figure 8.1: Feature Importance Scores of the Random Forest Model

8.4 Summary of Findings

The comparison showed that:

- Linear Regression slightly outperforms the Random Forest model across all evaluation metrics on the final cleaned dataset.
- This suggests that PM2.5 behavior in Calgary, as represented in the available features, is largely linear, particularly due to strong autoregressive signals.
- Autoregressive features (lag and rolling averages) remain the dominant predictors and play a central role in next-day PM2.5 forecasting.
- The improved cleaning and feature-engineering pipeline successfully preserved meaningful environmental patterns and enabled both models to perform reliably.

Based on these findings, the Linear Regression model provides the most accurate predictions for this dataset, while the Random Forest model serves as a strong non-linear alternative but does not offer additional accuracy improvements.

9. Results

This section summarizes the key outcomes of the PM2.5 forecasting project based on the cleaned dataset, exploratory analysis, and final modeling framework. The results focus on the environmental patterns identified in the data, the predictive relationships uncovered through feature engineering, and the practical implications of the model's outputs.

9.1 Key Environmental Patterns Identified

The final dataset revealed clear temporal and seasonal structures in Calgary's PM2.5 levels:

- Strong seasonality: PM2.5 concentrations consistently peaked during late summer and early fall, primarily coinciding with regional wildfire activity.
- Lowest pollution levels occurred during spring months, with relatively stable and low concentrations.
- Moderate winter elevations were observed, consistent with inversion events and reduced atmospheric mixing.

These patterns confirm that Calgary's air quality is heavily influenced by both seasonal cycles and episodic wildfire events.

9.2 Autoregressive Nature of PM2.5

The feature engineering results demonstrated that PM2.5 exhibits strong short-term persistence:

- The previous day's PM2.5 (lag-1) was one of the strongest predictors of the next day's concentration.
- The 3-day rolling average contributed additional stability, smoothing the short-term fluctuations while retaining major pollution trends.

This autoregressive behavior supports the use of short-term historical PM2.5 values as core components of forecasting models.

9.3 Influence of Meteorological Variables

Weather data played a meaningful but non-linear role in PM2.5 dynamics:

- Temperature showed an inverse relationship with PM2.5, indicating improved dispersion under warmer conditions.
- Precipitation displayed a mild cleaning effect, reducing particulate concentrations through wet deposition.
- Wind-related variables influenced pollutant transport and stagnant-air episodes, although the relationships were complex.

These relationships align with established atmospheric science and validate the inclusion of meteorological features.

9.4 Temporal Features and Weekly Cycles

The engineered temporal variables contributed essential contextual information:

- Month and season captured seasonal shifts in background pollutant levels.

- **Day of week (dow)** revealed slight differences between weekdays and weekends, suggesting the influence of human activity patterns such as commuting and industrial operations.

These findings show that PM2.5 levels are shaped by both natural cycles and socio-economic behavior.

9.5 Practical Implications of the Results

The aggregated results of the project demonstrate that:

- Short-term forecasting of PM2.5 in Calgary is feasible and accurate when combining autoregressive, meteorological, and temporal features.
- The modeling framework is well-suited for operational air-quality monitoring, supporting decision-making for public health protection.
- The identification of key predictors (lagged PM2.5, temperature, precipitation) provides actionable insights into the factors influencing pollution levels.

These results offer a strong foundation for further development of automated forecasting tools or real-time environmental monitoring systems.

9.6 Summary of Findings

The results of the analysis show that:

- PM2.5 concentrations in Calgary follow clear seasonal patterns, with elevated levels during late summer and early fall due to wildfire activity.
- Autoregressive features (lag-1 and rolling averages) provide the strongest predictive power, reflecting the temporal persistence of PM2.5.
- Meteorological variables contribute secondary but meaningful effects, influencing dispersion, stagnation, and scavenging processes.
- Both the Linear Regression and Random Forest models successfully capture these patterns, though the Linear Regression model achieves slightly better predictive performance on the final cleaned dataset.

Overall, the findings demonstrate that combining short-term historical PM2.5 values with essential meteorological and temporal features enables reliable next-day PM2.5 forecasting in Calgary. The results also highlight that, for this dataset, a simple linear model is sufficient to achieve strong predictive accuracy, while the Random Forest model remains a valuable non-linear alternative.

10. Discussion

The purpose of this section is to interpret the broader meaning of the project's findings, evaluate the strengths and limitations of the modeling approach, and outline the practical implications for air-quality forecasting in Calgary. While the previous sections presented the technical results, this discussion connects those outcomes to real-world environmental behavior and the overall effectiveness of the methodology.

10.1 Interpretation of Key Findings

The analysis revealed several important characteristics of Calgary's PM2.5 patterns:

- Clear seasonal cycles were observed: wildfire season (late summer to early fall) consistently produced the highest PM2.5 concentrations, while spring remained the period with the cleanest air.
- Strong autoregressive structure was identified, with lagged PM2.5 variables emerging as the most influential predictors. This demonstrates that air-quality conditions have short-term persistence driven by slow atmospheric changes and pollutant retention.
- Meteorological variables contributed non-linearly to PM2.5 levels. Temperature, precipitation, and wind-related variables influenced pollutant dispersion but did not follow simple linear relationships.

These insights reinforce that PM2.5 dynamics in Calgary are driven by a combination of environmental cycles, short-term pollutant memory, and episodic external events such as wildfires.

10.2 Strengths of the Modeling Approach

Several components of the modeling workflow contributed to the strong predictive performance:

- Comprehensive data cleaning (interpolation, outlier filtering, alignment) produced a stable and consistent dataset suitable for time-based forecasting.
- Feature engineering effectively captured critical temporal behaviors, especially the strong autoregressive structure of PM2.5.
- Evaluating both a simple linear model and a non-linear ensemble model provided insight into how Calgary's PM2.5 dynamics behave under different modeling assumptions.
- The Random Forest model offered interpretability through feature-importance analysis, helping identify the dominant predictors and their relative contributions.
- The overall workflow demonstrated robustness across typical and elevated pollution conditions, including wildfire-related peaks.

Together, these strengths demonstrate a reliable and practical framework for forecasting PM2.5 concentrations in an urban environment, while also highlighting that simpler models may perform just as well, or even better, when the underlying data relationships are largely linear.

10.3 Limitations

Despite its strong performance, the project has several inherent limitations:

- Lack of additional pollutant data: Only PM2.5 was modeled. Including co-pollutants such as NO₂ or O₃ may improve predictive depth.
- Absence of wildfire activity indicators: Although wildfire events strongly affect PM2.5, the model did not explicitly include variables representing fire intensity or smoke transport.
- Daily temporal resolution: Hourly data may reveal more detailed pollution dynamics but were not used in this study.
- Geographic constraints: The model was calibrated only for Calgary and may not generalize directly to other regions with different emission sources or weather patterns.

These limitations present clear opportunities for enhancing the forecasting system.

10.4 Practical Implications

The results of this study have meaningful implications for environmental monitoring and public health:

- Short-term air-quality forecasts can help residents plan outdoor activities, especially during smoke-filled periods.
- City agencies and health organizations may use such models to issue advisories earlier and more accurately.
- Predictive analytics can support resource allocation for vulnerable populations, such as distributing N95 masks during high-smoke days.
- The modeling workflow provides a scalable template that can be adapted to operational forecasting systems.

Overall, the study shows that data-driven modeling can serve as a valuable complement to traditional air-quality monitoring programs.

10.5 Future Work

Several enhancements could improve the performance and scope of the forecasting model:

- Incorporating hourly data, which may reveal intra-day pollution cycles
- Including wildfire-related variables, such as satellite-derived smoke plume density
- Testing advanced models (e.g., XGBoost, LSTM neural networks) to assess further improvement in accuracy
- Expanding the model to additional pollutants to build a more comprehensive air-quality prediction tool
- Deploying the model as an automated real-time forecasting system with continually updated inputs

These directions would strengthen the reliability, interpretability, and utility of short-term PM_{2.5} forecasting for Calgary and similar regions.

11. Conclusion

This project developed a data-driven framework for forecasting daily PM_{2.5} levels in Calgary using meteorological, temporal, and autoregressive features. The workflow, from data cleaning to feature engineering and model evaluation, showed that reliable short-term predictions can be achieved using publicly available environmental data.

The results highlighted clear seasonal pollution patterns, strong short-term persistence in PM_{2.5}, and meaningful contributions from meteorological variables. These findings informed the construction of an effective feature set for forecasting.

Model comparison showed that Linear Regression slightly outperformed the Random Forest Regressor, indicating that Calgary's PM_{2.5} dynamics are largely linear once autoregressive features are incorporated. Random Forest still provided valuable perspective through non-linear modeling and feature-importance insights.

Overall, the study demonstrates that a simple and interpretable model can produce accurate next-day PM_{2.5} forecasts for Calgary. Future work could include wildfire-specific indicators, hourly data, or more advanced algorithms to further enhance forecasting capability.

Appendix

This appendix provides supplemental information supporting the analysis, including data sources, preprocessing details, model hyperparameters, and full-resolution versions of all figures referenced in the report.

Appendix A. Data Sources

A.1 PM2.5 Data

National Air Pollution Surveillance (NAPS) Program

Hourly PM2.5 measurements used to compute daily averages.

A.2 Meteorological Data

Environment and Climate Change Canada (ECCC) – Historical Climate Data Portal

Daily weather observations including temperature, precipitation, wind, and atmospheric pressure.

Appendix B. Python Libraries

The following Python libraries were used for data cleaning, visualization, and model development:

- pandas
- numpy
- matplotlib / seaborn
- scikit-learn

These libraries enabled efficient and reliable implementation of the project workflow.

Appendix C. Feature Engineering Details

The following engineered variables were added to enhance predictive performance:

- PM25_lag1 – previous day's PM2.5
- PM25_roll3 – 3-day rolling average
- month – seasonal cycle indicator
- dow – day of week
- season – categorical seasonal grouping

Appendix D. Model Hyperparameters

D.1 Linear Regression Model

- Ordinary Least Squares (OLS)
- No regularization
- No scaling required

D.2 Random Forest Regressor

- n_estimators = 300
- max_depth = None
- random_state = 42
- n_jobs = -1

Appendix E. Exploratory Data Analysis (EDA) Figures

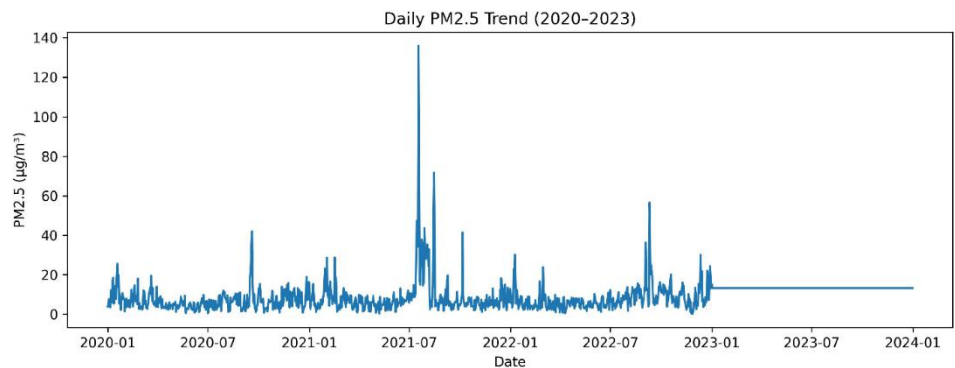


Figure E1. Daily PM2.5 Trend (2020–2023)
(Corresponds to Figure 5.1 in Section 5.1)

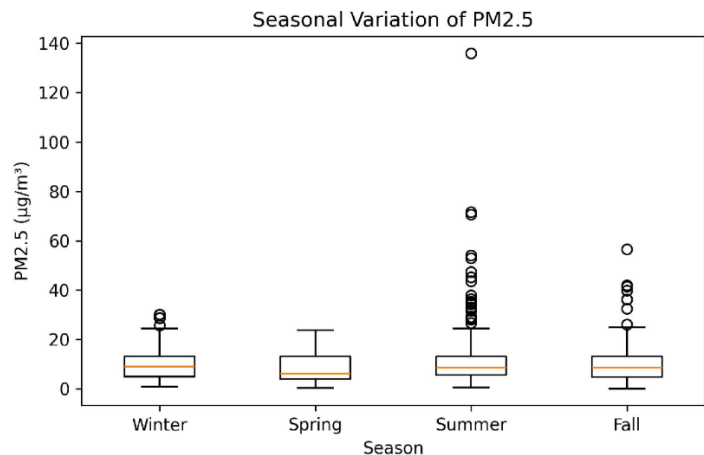


Figure E2. Seasonal Variation of PM2.5
(Corresponds to Figure 5.2 in Section 5.2)

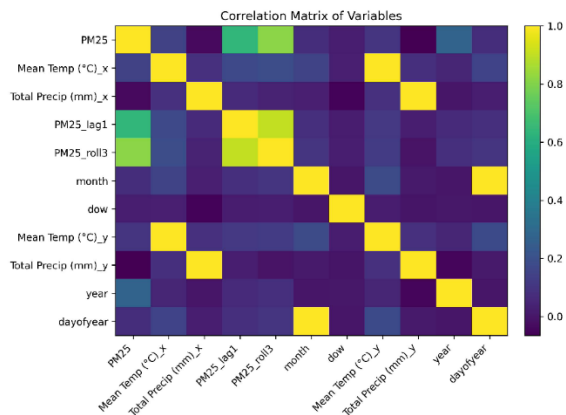


Figure E3. Correlation Matrix of Variables
(Corresponds to Figure 5.3 in Section 5.4)

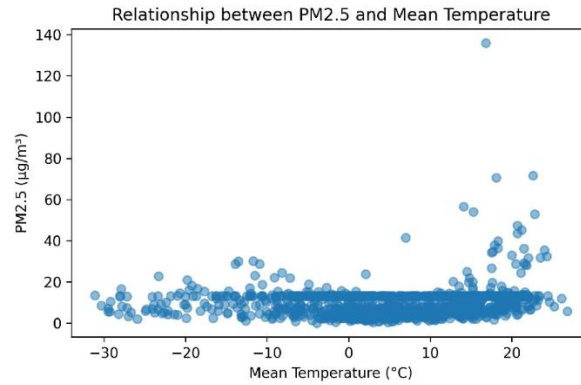


Figure E4. Scatter Plot: PM2.5 vs Mean Temperature
(Corresponds to Figure 5.4 in Section 5.4)

Appendix F. Model Evaluation Figures

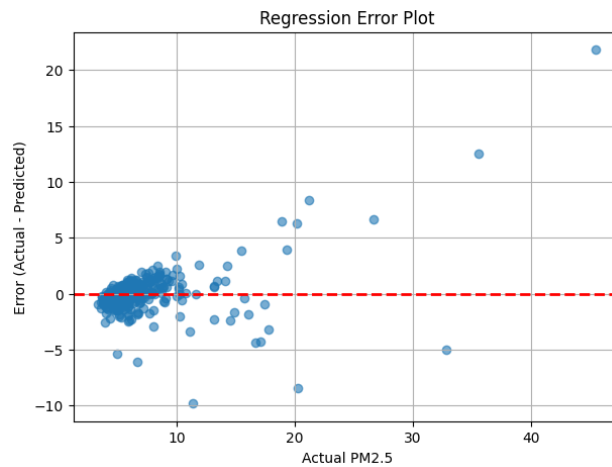


Figure F1. Regression Error Plot (Actual PM2.5 vs Prediction Error)
(Corresponds to Figure 7.1 in Section 7.3.1)

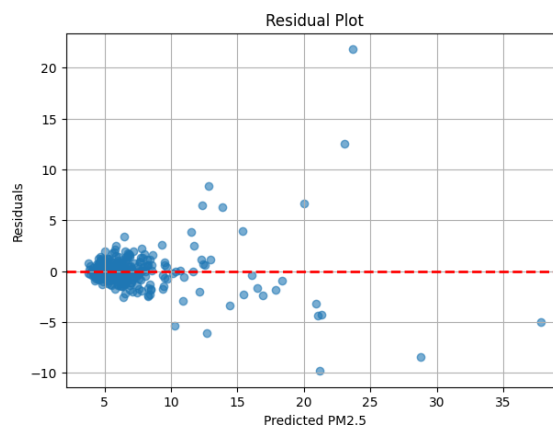


Figure F2. Residual Plot (Predicted PM2.5 vs Residuals)
(Corresponds to Figure 7.2 in Section 7.3.2)

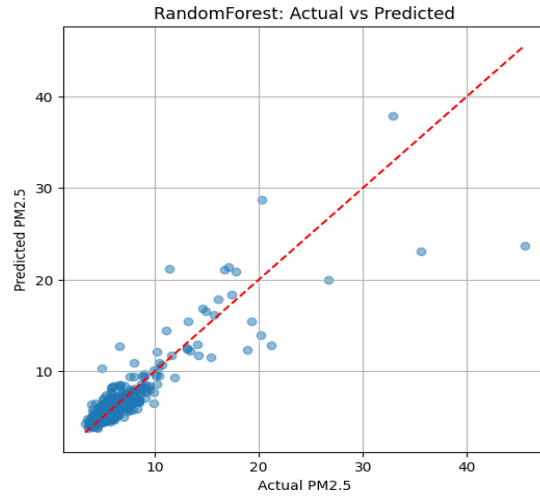


Figure F3. Actual vs Predicted PM2.5 Scatter Plot
(Corresponds to Figure 7.3 in Section 7.3.3)

Appendix G. Feature Importance Figure

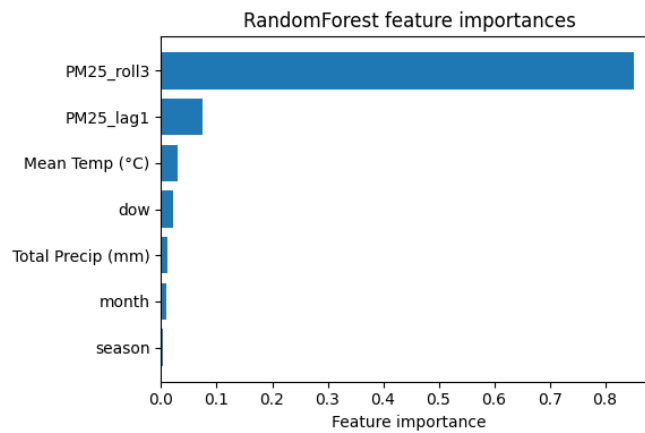


Figure G1. Random Forest Feature Importance Scores
(Corresponds to Figure 8.1 in Section 8.3)

Appendix H. Full Model Performance Metrics

H.1 Linear Regression

- MAE: 1.141
- RMSE: 2.068
- R^2 : 0.7969

H.2 Random Forest

- MAE: 1.152
- RMSE: 2.235
- R^2 : 0.7627

Appendix I. Data Cleaning and Model Code

Only the core sections of the code are provided in this appendix.

The full executable notebook is available in the accompanying project folder and GitHub repository.

I. Key Data Cleaning Code Snippet:

```
# Load raw PM2.5 and weather files

import pandas as pd

import glob

pm25_files = glob.glob("/content/drive/MyDrive/Raw_Data_PM25/PM25_*.csv")
weather_files = glob.glob("/content/drive/MyDrive/Raw_Data_PM25/en_climate_daily_*.csv")

df_pm25 = pd.concat([pd.read_csv(f) for f in pm25_files])
df_weather = pd.concat([pd.read_csv(f) for f in weather_files])

# Convert date column and aggregate hourly PM2.5 to daily means
df_pm25['date'] = pd.to_datetime(df_pm25['date'])
df_daily_pm25 = df_pm25.groupby('date')['PM25'].mean().reset_index()

# Merge PM2.5 with weather data
df_weather['date'] = pd.to_datetime(df_weather['date'])
df = pd.merge(df_daily_pm25, df_weather, on='date', how='inner')

# Handle missing values
df['Total Precip (mm)'] = df['Total Precip (mm)'].fillna(0)
df = df.interpolate(method='linear')

# Outlier removal (IQR method)
Q1 = df['PM25'].quantile(0.25)
Q3 = df['PM25'].quantile(0.75)
IQR = Q3 - Q1
df = df[(df['PM25'] >= Q1 - 1.5*IQR) & (df['PM25'] <= Q3 + 1.5*IQR)]

# Feature engineering
df['PM25_lag1'] = df['PM25'].shift(1)
df['PM25_rol3'] = df['PM25'].rolling(window=3).mean()
df['month'] = df['date'].dt.month
df['dow'] = df['date'].dt.dayofweek
df_cleaned = df.dropna().reset_index(drop=True)
```

I.2 Key Modeling Code Snippet:

```
# -----  
# Minimal PM2.5 Modeling Code (Appendix Version)  
# -----  
  
import pandas as pd  
import numpy as np  
from sklearn.model_selection import train_test_split  
from sklearn.linear_model import LinearRegression  
from sklearn.ensemble import RandomForestRegressor  
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score  
  
# Features and target  
X = df_model[["Mean Temp (°C)", "Total Precip (mm)",  
              "PM25_lag1", "PM25_roll3",  
              "month", "dow", "season"]]  
y = df_model["PM25"]  
  
# Train-test split  
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.2, random_state=42  
)  
  
# Linear Regression  
lr = LinearRegression().fit(X_train, y_train)  
y_pred_lr = lr.predict(X_test)  
  
# Random Forest  
rf = RandomForestRegressor(n_estimators=300, random_state=42).fit(X_train, y_train)  
y_pred_rf = rf.predict(X_test)  
  
# Metrics  
mae_lr = mean_absolute_error(y_test, y_pred_lr)  
rmse_lr = np.sqrt(mean_squared_error(y_test, y_pred_lr))  
r2_lr = r2_score(y_test, y_pred_lr)  
mae_rf = mean_absolute_error(y_test, y_pred_rf)  
rmse_rf = np.sqrt(mean_squared_error(y_test, y_pred_rf))  
r2_rf = r2_score(y_test, y_pred_rf)
```

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